Experiment 2, β-galactosidase kinetics: supplementary information and technical record

This document accompanies `experiment_2_submission.md` (the graded course report) and the journal-style version. It records the data audit, every analysis decision, the sensitivity analyses, the secondary analysis of the experimental dataset, and how to regenerate everything. All numbers are inserted from the processed outputs by `R/08_assemble_reports.R`; nothing is typed in by hand. Compiled 10 October 2026.

1. Datasets, provenance and terminology

Two datasets were analysed. The **reference dataset** is the set of course-provided measurements supplied by the laboratory instructor (Professor Samah Gadhoum; e-mail of 13 September 2026: “These are some results you can use for your results section. Please discuss your own results in the discussion part”) and was used for the primary kinetic analysis. The **experimental dataset** is the set of measurements collected by our laboratory group (14 GENESYS 10S printouts from 7 and 11 September 2026) and was analysed separately for technical interpretation and troubleshooting. The labels describe origin and analytical role; neither dataset is externally validated or published, and the experimental dataset is not a replicate of the reference dataset (different sessions, different operators). Original labels are retained in the raw files and in the crosswalk below.

Role	Dataset label in this project	Label on the source	File
reference	Reference Part I	Day1_PartI_ONP_and_enzyme_dilution.docx (tables 1-2)	data/raw/reference/part1_onp_spectrum.csv; part1_onp_standard_curve.csv
reference	Reference Part II	Day1_PartI_ONP_and_enzyme_dilution.docx (table 3)	data/raw/reference/part2_enzyme_dilution.csv
reference	Reference Part III (-IPTG)	Day2 pptx / instructor_day2_primary_results.csv: ‘No IPTG’	data/raw/reference/part3_substrate_series.csv
reference	Reference Part III (+IPTG)	Day2 pptx / instructor_day2_primary_results.csv: ‘+ IPTG’	data/raw/reference/part3_substrate_series.csv
experimental	Experimental Part II	printouts labelled ‘Day 1’ (red ink), 7 Sep 2026	data/raw/experimental/absorbance_raw.csv
experimental	Experimental Part III (-IPTG)	printouts labelled ‘Day 2 Blue’, 11 Sep 2026	data/raw/experimental/absorbance_raw.csv
experimental	Experimental Part III (+IPTG)	printouts labelled ‘Day 2 Black’, 11 Sep 2026	data/raw/experimental/absorbance_raw.csv

Table A0. Dataset crosswalk: labels used in this project, labels on the source files, and file locations.

2. Data audit

2.1 Reference files against the transcribed CSVs

All 17 spectrum values and all 7 standard-curve values in the supplied `instructor_part1_values.csv` match the course .docx; all 60 Part III values in `instructor_day2_primary_results.csv` match the .pptx. The supplied Part I CSV omits the enzyme-dilution table that the .docx contains (tubes A to E, relative enzyme 1.00 to 0.15, five time points); it was transcribed from the .docx into `part2_enzyme_dilution.csv`. The pipeline re-verifies every transcription against the source files when they are present (`data/processed/validation_report.csv`).

2.2 Standard-curve tube assignment

The manual describes eight preparations (a blank and standards 1 to 7 with 0.25, 0.50, 0.75, 1.0, 1.3, 1.6 and 2.0 mL of 1 mM ONP in 3.0 mL). The course table lists seven readings labelled tubes 1 to 7 (0, 0.428, 0.856, 1.240, 1.723, 2.256, 3.236). Classmates queried the discrepancy and the instructor clarified that the first reading is the zero-concentration point, listed against the first reported tube rather than as a separately numbered blank. Two facts fix the assignment of the remaining six readings: a 0.25 mL standard (0.083 mM) could not read 0, and the spectrum tube (manual standard 4, 1.0 mL) read 1.723 at 415 nm, identical to the fifth reported reading. Under the adopted mapping (reported tube k = manual standard k − 1) the absorbance-to-concentration ratios of the first five standards are 4.96 to 5.21 per mM, as Beer–Lambert behaviour requires; under the alternative (reported tubes 2 to 7 = manual standards 2 to 7) the ratios would fall from 2.57 to 4.85 per mM and the spectrum tube would not match. The manual’s standard 7 (0.667 mM) is therefore the preparation with no reported reading; whether it was not prepared or not reported cannot be determined from the files, and no absorbance was assumed for it. The zero-concentration preparation is chemically the blank (2.0 mL Z buffer + 1.0 mL Na_2CO_3 read against itself), which is why it reads 0; it is shown on the standard curve but not used in the fit.

Reported tube (course table)	Manual preparation	ONP 1 mM (mL)	Z buffer (mL)	Na2CO3 1 M (mL)	[ONP] in cuvette (mM)	A415 (reported)	Basis of the mapping
1	Blank (0 mL ONP)	0	2	1	0	0.000	Reported as tube 1 with OD 0; the instructor confirmed it is the zero-concentration point (blank-like preparation, read against itself)
2	Standard 1	0.25	1.75	1	0.0833	0.428	Next reported reading; mapping consistent with Beer-Lambert ratio A/[ONP] = 5.1 per mM
3	Standard 2	0.5	1.5	1	0.1667	0.856	A/[ONP] = 5.1 per mM
4	Standard 3	0.75	1.25	1	0.25	1.240	A/[ONP] = 5.0 per mM
5	Standard 4	1	1	1	0.3333	1.723	Identical to the spectrum reading of manual tube 4 at 415 nm (1.723), which anchors the mapping
6	Standard 5	1.3	0.7	1	0.4333	2.256	A/[ONP] = 5.2 per mM
7	Standard 6	1.6	0.4	1	0.5333	3.236	A/[ONP] = 6.1 per mM; highest reported reading
not reported	Standard 7	2	0	1	0.6667	none	Not in the reported table (0.667 mM); unknown whether this standard was not prepared or not reported

Table A1. Mapping of the reported standard-curve readings to the manual’s preparations.

2.3 Features of the reference progress curves

Enzyme-dilution tube A has an 8-min reading (0.965) far below its 2 to 6 min trend, with the 10-min reading back on the line; tube C is similar. Uninhibited tubes B to F decelerate from the first interval onward: the second increment is 50 to 73% of the first and the fourth is 6 to 50% of the first. Inhibited tube A has an irregular 4-min reading (0.480); inhibited tubes B to F are close to linear with small increments (0.005 to 0.07 A per 2 min), where the 0.001 A read-out resolution matters.

dataset	condition_label	tube	observed_time_points	n_missing	increments	inc2_over_inc1	last_over_prior	source_check	observation
Reference Part II	Enzyme dilution (Part II)	A	2, 4, 6, 8, 10	0	0.270 / 0.274 / 0.089 / 0.442	1.01	2.09	Transcribed from the course .docx (table absent from the supplied CSV)	8-min reading (0.965) far below the 2-6 min trend; 10-min reading returns to it
Reference Part II	Enzyme dilution (Part II)	B	2, 4, 6, 8, 10	0	0.242 / 0.183 / 0.220 / 0.194	0.756	0.902	Transcribed from the course .docx (table absent from the supplied CSV)	Approximately linear; small increments (read-out resolution 0.001 A matters)
Reference Part II	Enzyme dilution (Part II)	C	2, 4, 6, 8, 10	0	0.150 / 0.178 / 0.093 / 0.144	1.19	1.03	Transcribed from the course .docx (table absent from the supplied CSV)	8-min reading below trend
Reference Part II	Enzyme dilution (Part II)	D	2, 4, 6, 8, 10	0	0.040 / 0.057 / 0.062 / 0.033	1.43	0.623	Transcribed from the course .docx (table absent from the supplied CSV)	Tubes D (0.25) and E (0.15 relative enzyme) gave nearly equal rates
Reference Part II	Enzyme dilution (Part II)	E	2, 4, 6, 8, 10	0	0.047 / 0.048 / 0.040 / 0.050	1.02	1.11	Transcribed from the course .docx (table absent from the supplied CSV)	Tubes D (0.25) and E (0.15 relative enzyme) gave nearly equal rates
Reference Part III (+IPTG)	ONPG series + IPTG	A	2, 4, 6, 8, 10	0	0.282 / 0.106 / 0.198 / 0.182	0.376	0.932	Supplied CSV matches the course .pptx (all 30 values per condition)	Irregular 4-min reading (0.480); later increments below the first
Reference Part III (+IPTG)	ONPG series + IPTG	B	2, 4, 6, 8, 10	0	0.062 / 0.069 / 0.043 / 0.055	1.11	0.948	Supplied CSV matches the course .pptx (all 30 values per condition)	Slows after 6 min
Reference Part III (+IPTG)	ONPG series + IPTG	C	2, 4, 6, 8, 10	0	0.041 / 0.042 / 0.026 / 0.033	1.02	0.908	Supplied CSV matches the course .pptx (all 30 values per condition)	Slows after 6 min
Reference Part III (+IPTG)	ONPG series + IPTG	D	2, 4, 6, 8, 10	0	0.022 / 0.017 / 0.017 / 0.014	0.773	0.75	Supplied CSV matches the course .pptx (all 30 values per condition)	Approximately linear; small increments (read-out resolution 0.001 A matters)
Reference Part III (+IPTG)	ONPG series + IPTG	E	2, 4, 6, 8, 10	0	0.015 / 0.012 / 0.013 / 0.009	0.8	0.675	Supplied CSV matches the course .pptx (all 30 values per condition)	Approximately linear; small increments (read-out resolution 0.001 A matters)
Reference Part III (+IPTG)	ONPG series + IPTG	F	2, 4, 6, 8, 10	0	0.008 / 0.006 / 0.006 / 0.005	0.75	0.75	Supplied CSV matches the course .pptx (all 30 values per condition)	Approximately linear; small increments (read-out resolution 0.001 A matters)
Reference Part III (-IPTG)	ONPG series, no inhibitor	A	2, 4, 6, 8, 10	0	0.264 / 0.269 / 0.279 / 0.234	1.02	0.865	Supplied CSV matches the course .pptx (all 30 values per condition)	Linear to 10 min
Reference Part III (-IPTG)	ONPG series, no inhibitor	B	2, 4, 6, 8, 10	0	0.181 / 0.133 / 0.112 / 0.095	0.735	0.669	Supplied CSV matches the course .pptx (all 30 values per condition)	Rate falls from the first interval onward (increments decline monotonically)
Reference Part III (-IPTG)	ONPG series, no inhibitor	C	2, 4, 6, 8, 10	0	0.122 / 0.079 / 0.068 / 0.043	0.648	0.48	Supplied CSV matches the course .pptx (all 30 values per condition)	Rate falls from the first interval onward (increments decline monotonically)
Reference Part III (-IPTG)	ONPG series, no inhibitor	D	2, 4, 6, 8, 10	0	0.062 / 0.039 / 0.029 / 0.061	0.629	1.41	Supplied CSV matches the course .pptx (all 30 values per condition)	Rate falls from the first interval onward (increments decline monotonically)
Reference Part III (-IPTG)	ONPG series, no inhibitor	E	2, 4, 6, 8, 10	0	0.032 / 0.018 / 0.005 / 0.009	0.563	0.491	Supplied CSV matches the course .pptx (all 30 values per condition)	Rate falls from the first interval onward (increments decline monotonically)
Reference Part III (-IPTG)	ONPG series, no inhibitor	F	2, 4, 6, 8, 10	0	0.020 / 0.010 / 0.007 / 0.006	0.5	0.486	Supplied CSV matches the course .pptx (all 30 values per condition)	Rate falls from the first interval onward (increments decline monotonically)

Table A2. Audit of the reference progress curves.

2.4 Experimental printouts

All 14 photographs were compared with the transcription; every legible value matches (one last digit, Part III +IPTG tube A at 6 min, is under glare in the photograph and reads 0.47x; the recorded 0.478 sits on that tube's straight line). Part III +IPTG tube E has no data (sample discarded before measurement) and is `NA` throughout. Part III +IPTG tube B (nominally 1.10 mM ONPG) is anomalous as a whole curve: increments 0.237, 0.242, 0.247 over the first 8 min exceed those of tube A (0.157, 0.158, 0.150) at twice the substrate, and its 10-min increment (0.111) is 46% of the preceding mean. The mapping of the blue- and black-labelled printouts to the uninhibited and inhibited series was established from tube count, the manual's enzyme-standardisation specification and the rate structure (blue A reached 1.061 at 10 min and black A only 0.780; black rates fell 4.3-fold from A to D against 1.3-fold for blue); the reference dataset shows the same pattern (inhibited A 0.966, uninhibited A 1.328 at 10 min).

dataset	source_label	condition_label	tube	observed_time_points	n_missing	increments	inc2_over_inc1	last_over_prior	verification	instrument_clock	observation
Experimental Part II	Day 1	Enzyme dilution (Part II)	A	2, 4, 6, 8, 10	0	0.268 / 0.273 / 0.320 / 0.191	1.02	0.666	all five values match CSV	9:45pm 7Sep26	Linear to 8 or 10 min
Experimental Part II	Day 1	Enzyme dilution (Part II)	B	2, 4, 6, 8, 10	0	0.174 / 0.179 / 0.161 / 0.166	1.03	0.969	all five values match CSV (photo was taken upside down; the file is stored rotated 180° for legibility)	9:54pm 7Sep26	Linear to 8 or 10 min
Experimental Part II	Day 1	Enzyme dilution (Part II)	C	2, 4, 6, 8, 10	0	0.122 / 0.120 / 0.124 / 0.084	0.984	0.689	all five values match CSV	9:58pm 7Sep26	Linear to 8 or 10 min
Experimental Part II	Day 1	Enzyme dilution (Part II)	D	2, 4, 6, 8, 10	0	0.056 / 0.055 / 0.059 / 0.048	0.982	0.847	all five values match CSV	10:04pm 7Sep26	Linear to 8 or 10 min
Experimental Part II	Day 1	Enzyme dilution (Part II)	E	2, 4, 6, 8, 10	0	0.043 / 0.037 / 0.042 / 0.036	0.86	0.885	all five values match CSV	10:07pm 7Sep26	Linear to 8 or 10 min
Experimental Part III (+IPTG)	Day 2 Black	ONPG series + IPTG	A	2, 4, 6, 8, 10	0	0.157 / 0.158 / 0.150 / 0.152	1.01	0.981	four values match CSV; ID 3 legible as 0.47x with last digit under glare, consistent with CSV 0.478 but not independently confirmable from the photo	4:29am 11Sep26	ID 3 last digit under photo glare; CSV 0.478 consistent with legible 0.47x
Experimental Part III (+IPTG)	Day 2 Black	ONPG series + IPTG	B	2, 4, 6, 8, 10	0	0.237 / 0.242 / 0.247 / 0.111	1.02	0.459	all five values match CSV	4:25am 11Sep26	Whole curve faster than tube A at half the nominal ONPG; 10-min value 1.106 retained (46% of predicted increment)
Experimental Part III (+IPTG)	Day 2 Black	ONPG series + IPTG	C	2, 4, 6, 8, 10	0	0.078 / 0.086 / 0.060 / 0.058	1.1	0.777	all five values match CSV	4:33am 11Sep26	Linear to 8 or 10 min
Experimental Part III (+IPTG)	Day 2 Black	ONPG series + IPTG	D	2, 4, 6, 8, 10	0	0.041 / 0.034 / 0.035 / 0.037	0.829	1.01	all five values match CSV	4:20am 11Sep26	Linear to 8 or 10 min
Experimental Part III (+IPTG)	Day 2 Black	ONPG series + IPTG	E	none	5	NA	NA	NA	no printout: sample discarded before measurement	NA	All five observations missing: sample discarded before measurement (kept NA)
Experimental Part III (-IPTG)	Day 2 Blue	ONPG series, no inhibitor	A	2, 4, 6, 8, 10	0	0.221 / 0.216 / 0.242 / 0.123	0.977	0.543	all five values match CSV	3:32am 11Sep26	Linear to 8 or 10 min
Experimental Part III (-IPTG)	Day 2 Blue	ONPG series, no inhibitor	B	2, 4, 6, 8, 10	0	0.212 / 0.219 / 0.209 / 0.201	1.03	0.942	all five values match CSV	3:36am 11Sep26	Linear to 8 or 10 min
Experimental Part III (-IPTG)	Day 2 Blue	ONPG series, no inhibitor	C	2, 4, 6, 8, 10	0	0.188 / 0.161 / 0.193 / 0.138	0.856	0.764	all five values match CSV	3:39am 11Sep26	Linear to 8 or 10 min
Experimental Part III (-IPTG)	Day 2 Blue	ONPG series, no inhibitor	D	2, 4, 6, 8, 10	0	0.184 / 0.165 / 0.123 / 0.090	0.897	0.572	all five values match CSV	3:46am 11Sep26	Decelerates after 6 min

dataset	source_label	condition_label	tube	observed_time_points	n_missing	increments	inc2_over_inc1	last_over_prior	verification	instrument_clock	observation
Experimental Part III (-IPTG)	Day 2 Blue	ONPG series, no inhibitor	E	2, 4, 6, 8, 10	0	0.090 / 0.048 / 0.052 / 0.045	0.533	0.711	all five values match CSV	3:51am 11Sep26	Non-linear from the outset (2nd increment 53% of 1st)

Table A3. Audit of the experimental progress curves.

3. Methods of analysis

3.1 Concentrations and volumes

[ONPG] = 3.3 mM × V_ONPG/12.0 mL (2.20, 1.10, 0.55, 0.275, 0.1375, 0.06875 mM for tubes A to F); [IPTG] = 5.5 mM × 1.0/12.0 = 0.4583 mM in every inhibited tube; ONP standards 1 mM × V/3.0 mL. Every reaction tube in the manual sums to 12.0 mL and every Part I preparation to 3.0 mL (checked programmatically; `validation_report.csv`).

Part	Condition	Tube	Z buffer (mL)	ONPG 3.3 mM (mL)	IPTG 5.5 mM (mL)	Enzyme (mL)	[ONPG] in reaction (mM)	[IPTG] in reaction (mM)	Relative enzyme
Part II	Enzyme dilution (Part II)	A	3.8	8	0	0.2	2.2	0	1
Part II	Enzyme dilution (Part II)	B	3.85	8	0	0.15	2.2	0	0.75
Part II	Enzyme dilution (Part II)	C	3.9	8	0	0.1	2.2	0	0.5
Part II	Enzyme dilution (Part II)	D	3.95	8	0	0.05	2.2	0	0.25
Part II	Enzyme dilution (Part II)	E	3.97	8	0	0.03	2.2	0	0.15
Part III	ONPG series, no inhibitor	A	3.8	8	0	0.2	2.2	0	NA
Part III	ONPG series, no inhibitor	B	7.8	4	0	0.2	1.1	0	NA
Part III	ONPG series, no inhibitor	C	9.8	2	0	0.2	0.55	0	NA
Part III	ONPG series, no inhibitor	D	10.8	1	0	0.2	0.275	0	NA
Part III	ONPG series, no inhibitor	E	11.3	0.5	0	0.2	0.1375	0	NA
Part III	ONPG series, no inhibitor	F	11.55	0.25	0	0.2	0.0687	0	NA
Part III	ONPG series + IPTG	A	2.8	8	1	0.2	2.2	0.4583	NA
Part III	ONPG series + IPTG	B	6.8	4	1	0.2	1.1	0.4583	NA
Part III	ONPG series + IPTG	C	8.8	2	1	0.2	0.55	0.4583	NA
Part III	ONPG series + IPTG	D	9.8	1	1	0.2	0.275	0.4583	NA
Part III	ONPG series + IPTG	E	10.3	0.5	1	0.2	0.1375	0.4583	NA
Part III	ONPG series + IPTG	F	10.55	0.25	1	0.2	0.0687	0.4583	NA

Table A4. Tube compositions from the manual with derived concentrations.

3.2 Rule for the initial velocity

The manual asks for V_0 from the initial linear portion of each progress curve, with the points used identified. The rule applies that instruction identically to every curve. Tier 1: fit $A_{415} = a + bt$ to the 2, 4 and 6 min points and extend one point at a time while the added increment lies within 20% (or 0.010 A, whichever larger; the absolute floor protects low-signal curves from read-out noise) of the increment predicted by the current line; $V_0 = b$. Tier 2: when the second and third increments both fall below the first by more than the tolerance, there is no initial linear portion within the sampled interval and a regression on 2 to 6 min would average over a falling rate; V_0 is taken from the earliest interval available, the 0 to 2 min secant through the origin, and flagged. Tier 2 applied to 6 of the 17 reference curves (uninhibited B to F and inhibited A) and to one experimental curve (uninhibited E). The rule depends only on the shape of each curve, not on the kinetic parameters that result from it.

Origin assumption. No reading exists at t = 0, so the Tier-2 secant assumes zero absorbance at t = 0. The regression intercepts of the 24 curves that were linear estimate the actual offset (reaction before the timer, the matrix difference between the stopped sample and the Z-buffer blank, read-out offset): median 0.013 A, range -0.020 to 0.059 A (reference curves alone: median 0.014 A, maximum 0.059 A). A positive offset makes the secant overstate the initial rate by offset/2 min, i.e. typically 0.006 and at most 0.030 ΔA_{415} min^{-1}. The “lower bound” label on secant estimates therefore holds only if the offset is zero; estimator O below carries the median correction through every kinetic parameter.

Dataset	Linear curves (n)	Median intercept (A415)	Min	Max	Median secant bias (dA415/min)	Max secant bias (dA415/min)
Experimental dataset (our laboratory group)	13	0.0112	-0.0205	0.0307	0.0056	0.0153
Reference dataset (course-provided)	11	0.0142	-0.0017	0.0593	0.0071	0.0297
both datasets	24	0.0127	-0.0205	0.0593	0.0064	0.0297

Table A5. Intercepts of the linear curves and the implied bias of the 0 to 2 min secant.

Dataset	Tube	Concentration variable	Increments (A415 per 2 min)	V0 estimator	Points used (min)	V0 (dA415/min)	Regression R2	Intercept (A415)	Notes
Reference Part II	A	relative enzyme 1.00	0.270 / 0.274 / 0.089 / 0.442	regression 2-6 min	2, 4, 6	0.136	1	0.0593	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Reference Part II	B	relative enzyme 0.75	0.242 / 0.183 / 0.220 / 0.194	regression 2-10 min	2, 4, 6, 8, 10	0.1041	0.9983	0.0509	All five points within the initial linear window

Dataset	Tube	Concentration variable	Increments (A415 per 2 min)	V0 estimator	Points used (min)	V0 (dA415/min)	Regression R2	Intercept (A415)	Notes
Reference Part II	C	relative enzyme 0.50	0.150 / 0.178 / 0.093 / 0.144	regression 2-6 min	2, 4, 6	0.082	0.9976	0.0023	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Reference Part II	D	relative enzyme 0.25	0.040 / 0.057 / 0.062 / 0.033	regression 2-6 min	2, 4, 6	0.0243	0.9899	0.0187	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Reference Part II	E	relative enzyme 0.15	0.047 / 0.048 / 0.040 / 0.050	regression 2-10 min	2, 4, 6, 8, 10	0.0229	0.9991	0.018	All five points within the initial linear window
Reference Part III (+IPTG)	A	[ONPG] 2.2000 mM + [IPTG] 0.458 mM	0.282 / 0.106 / 0.198 / 0.182	0-2 min secant (no initial linear portion)	0 (origin), 2	0.099	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)
Reference Part III (+IPTG)	B	[ONPG] 1.1000 mM + [IPTG] 0.458 mM	0.062 / 0.069 / 0.043 / 0.055	regression 2-6 min	2, 4, 6	0.0328	0.999	-0.0017	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Reference Part III (+IPTG)	C	[ONPG] 0.5500 mM + [IPTG] 0.458 mM	0.041 / 0.042 / 0.026 / 0.033	regression 2-6 min	2, 4, 6	0.0208	1	0.0033	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Reference Part III (+IPTG)	D	[ONPG] 0.2750 mM + [IPTG] 0.458 mM	0.022 / 0.017 / 0.017 / 0.014	regression 2-10 min	2, 4, 6, 8, 10	0.0087	0.9936	0.0142	All five points within the initial linear window
Reference Part III (+IPTG)	E	[ONPG] 0.1375 mM + [IPTG] 0.458 mM	0.015 / 0.012 / 0.013 / 0.009	regression 2-10 min	2, 4, 6, 8, 10	0.0062	0.9935	0.0093	All five points within the initial linear window
Reference Part III (+IPTG)	F	[ONPG] 0.0687 mM + [IPTG] 0.458 mM	0.008 / 0.006 / 0.006 / 0.005	regression 2-10 min	2, 4, 6, 8, 10	0.0031	0.9928	0.0058	All five points within the initial linear window
Reference Part III (-IPTG)	A	[ONPG] 2.2000 mM	0.264 / 0.269 / 0.279 / 0.234	regression 2-10 min	2, 4, 6, 8, 10	0.132	0.9993	0.021	All five points within the initial linear window
Reference Part III (-IPTG)	B	[ONPG] 1.1000 mM	0.181 / 0.133 / 0.112 / 0.095	0-2 min secant (no initial linear portion)	0 (origin), 2	0.101	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)
Reference Part III (-IPTG)	C	[ONPG] 0.5500 mM	0.122 / 0.079 / 0.068 / 0.043	0-2 min secant (no initial linear portion)	0 (origin), 2	0.078	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)
Reference Part III (-IPTG)	D	[ONPG] 0.2750 mM	0.062 / 0.039 / 0.029 / 0.061	0-2 min secant (no initial linear portion)	0 (origin), 2	0.055	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)
Reference Part III (-IPTG)	E	[ONPG] 0.1375 mM	0.032 / 0.018 / 0.005 / 0.009	0-2 min secant (no initial linear portion)	0 (origin), 2	0.058	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)
Reference Part III (-IPTG)	F	[ONPG] 0.0687 mM	0.020 / 0.010 / 0.007 / 0.006	0-2 min secant (no initial linear portion)	0 (origin), 2	0.0235	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)

Table A6. Initial velocities of the reference curves (primary data).

Dataset	Tube	Concentration variable	Increments (A415 per 2 min)	V0 estimator	Points used (min)	V0 (dA415/min)	Regression R2	Intercept (A415)	Notes
Experimental Part II	A	relative enzyme 1.00	0.268 / 0.273 / 0.320 / 0.191	regression 2-8 min	2, 4, 6, 8	0.1428	0.9981	-0.0205	Point(s) at 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part II	B	relative enzyme 0.75	0.174 / 0.179 / 0.161 / 0.166	regression 2-10 min	2, 4, 6, 8, 10	0.085	0.9996	0.0142	All five points within the initial linear window
Experimental Part II	C	relative enzyme 0.50	0.122 / 0.120 / 0.124 / 0.084	regression 2-8 min	2, 4, 6, 8	0.0609	1	0.011	Point(s) at 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part II	D	relative enzyme 0.25	0.056 / 0.055 / 0.059 / 0.048	regression 2-10 min	2, 4, 6, 8, 10	0.0275	0.9991	0.011	All five points within the initial linear window
Experimental Part II	E	relative enzyme 0.15	0.043 / 0.037 / 0.042 / 0.036	regression 2-10 min	2, 4, 6, 8, 10	0.0197	0.9992	0.0091	All five points within the initial linear window
Experimental Part III (+IPTG)	A	[ONPG] 2.2000 mM + [IPTG] 0.458 mM	0.157 / 0.158 / 0.150 / 0.152	regression 2-10 min	2, 4, 6, 8, 10	0.0771	0.9999	0.0112	All five points within the initial linear window
Experimental Part III (+IPTG)	B	[ONPG] 1.1000 mM + [IPTG] 0.458 mM	0.237 / 0.242 / 0.247 / 0.111	regression 2-8 min	2, 4, 6, 8	0.121	0.9999	0.0245	10-min point excluded (46% of predicted increment); whole curve faster than tube A, see audit

Dataset	Tube	Concentration variable	Increments (A415 per 2 min)	V0 estimator	Points used (min)	V0 (dA415/min)	Regression R2	Intercept (A415)	Notes
Experimental Part III (+IPTG)	C	[ONPG] 0.5500 mM + [IPTG] 0.458 mM	0.078 / 0.086 / 0.060 / 0.058	regression 2-6 min	2, 4, 6	0.041	0.9992	-0.0033	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part III (+IPTG)	D	[ONPG] 0.2750 mM + [IPTG] 0.458 mM	0.041 / 0.034 / 0.035 / 0.037	regression 2-10 min	2, 4, 6, 8, 10	0.0181	0.9991	0.0067	All five points within the initial linear window
Experimental Part III (+IPTG)	E	[ONPG] 0.1375 mM + [IPTG] 0.458 mM	NA	none	none	NA	NA	NA	No V0: sample discarded before measurement (all observations NA)
Experimental Part III (-IPTG)	A	[ONPG] 2.2000 mM	0.221 / 0.216 / 0.242 / 0.123	regression 2-8 min	2, 4, 6, 8	0.1126	0.9994	0.03	Point(s) at 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part III (-IPTG)	B	[ONPG] 1.1000 mM	0.212 / 0.219 / 0.209 / 0.201	regression 2-10 min	2, 4, 6, 8, 10	0.1055	0.9998	0.0158	All five points within the initial linear window
Experimental Part III (-IPTG)	C	[ONPG] 0.5500 mM	0.188 / 0.161 / 0.193 / 0.138	regression 2-8 min	2, 4, 6, 8	0.0894	0.9989	0.023	Point(s) at 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part III (-IPTG)	D	[ONPG] 0.2750 mM	0.184 / 0.165 / 0.123 / 0.090	regression 2-6 min	2, 4, 6	0.0872	0.999	0.0307	Point(s) at 8, 10 min departed from the initial trend and were excluded from the regression; raw values retained
Experimental Part III (-IPTG)	E	[ONPG] 0.1375 mM	0.090 / 0.048 / 0.052 / 0.045	0-2 min secant (no initial linear portion)	0 (origin), 2	0.0555	NA	NA	Second and third increments both below the first by more than the tolerance: no initial linear portion; 0-2 min secant used (lower bound if the t = 0 offset is zero)

Table A6b. Initial velocities of the experimental curves.

Role	Dataset	Tube	Window	n	Slope (dA415/min)	Intercept	R2	RMSE	Selected
Experimental dataset (our laboratory group)	Experimental Part II	A	2-6 min	3	0.1353	0.0047	1	0.002	
Experimental dataset (our laboratory group)	Experimental Part II	A	2-8 min	4	0.1428	-0.0205	0.9981	0.0195	yes
Experimental dataset (our laboratory group)	Experimental Part II	A	2-10 min	5	0.1348	0.0113	0.9955	0.0331	
Experimental dataset (our laboratory group)	Experimental Part II	B	2-6 min	3	0.0883	0.0027	0.9999	0.002	
Experimental dataset (our laboratory group)	Experimental Part II	B	2-8 min	4	0.086	0.01	0.9995	0.0059	
Experimental dataset (our laboratory group)	Experimental Part II	B	2-10 min	5	0.085	0.0142	0.9996	0.0061	yes
Experimental dataset (our laboratory group)	Experimental Part II	C	2-6 min	3	0.0605	0.0123	1	0.0008	
Experimental dataset (our laboratory group)	Experimental Part II	C	2-8 min	4	0.0609	0.011	1	0.0012	yes
Experimental dataset (our laboratory group)	Experimental Part II	C	2-10 min	5	0.0572	0.0258	0.9958	0.0135	
Experimental dataset (our laboratory group)	Experimental Part II	D	2-6 min	3	0.0277	0.0097	1	0.0004	
Experimental dataset (our laboratory group)	Experimental Part II	D	2-8 min	4	0.0282	0.008	0.9998	0.0013	
Experimental dataset (our laboratory group)	Experimental Part II	D	2-10 min	5	0.0275	0.011	0.9991	0.0029	yes
Experimental dataset (our laboratory group)	Experimental Part II	E	2-6 min	3	0.02	0.008	0.9981	0.0024	
Experimental dataset (our laboratory group)	Experimental Part II	E	2-8 min	4	0.0202	0.0075	0.9992	0.0018	
Experimental dataset (our laboratory group)	Experimental Part II	E	2-10 min	5	0.0197	0.0091	0.9992	0.0021	yes
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	A	2-6 min	3	0.0788	0.0053	1	0.0004	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	A	2-8 min	4	0.0776	0.009	0.9999	0.0029	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	A	2-10 min	5	0.0771	0.0112	0.9999	0.0031	yes
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	B	2-6 min	3	0.1198	0.0287	1	0.002	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	B	2-8 min	4	0.121	0.0245	0.9999	0.0035	yes
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	B	2-10 min	5	0.1082	0.0759	0.986	0.047	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	C	2-6 min	3	0.041	-0.0033	0.9992	0.0033	yes
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	C	2-8 min	4	0.0379	0.007	0.9952	0.0083	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	C	2-10 min	5	0.0355	0.0166	0.9927	0.0111	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	D	2-6 min	3	0.0187	0.0047	0.9971	0.0029	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	D	2-8 min	4	0.0182	0.0065	0.9982	0.0025	
Experimental dataset (our laboratory group)	Experimental Part III (+IPTG)	D	2-10 min	5	0.0181	0.0067	0.9991	0.002	yes
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	A	2-6 min	3	0.1092	0.0413	1	0.002	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	A	2-8 min	4	0.1126	0.03	0.9994	0.0089	yes
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	A	2-10 min	5	0.1031	0.0682	0.9911	0.0356	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	B	2-6 min	3	0.1077	0.0073	0.9999	0.0029	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	B	2-8 min	4	0.1069	0.01	0.9999	0.0029	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	B	2-10 min	5	0.1055	0.0158	0.9998	0.0058	yes
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	C	2-6 min	3	0.0873	0.03	0.998	0.011	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	C	2-8 min	4	0.0894	0.023	0.9989	0.0095	yes
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	C	2-10 min	5	0.0857	0.0376	0.9976	0.0154	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	D	2-6 min	3	0.0872	0.0307	0.999	0.0078	yes
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	D	2-8 min	4	0.0791	0.058	0.9924	0.0219	
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	D	2-10 min	5	0.0706	0.0918	0.9812	0.0357	

Role	Dataset	Tube	Window	n	Slope (dA415/min)	Intercept	R2	RMSE	Selected
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	E	2-6 min	3	0.0345	0.049	0.97	0.0171	secant used instead
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	E	2-8 min	4	0.0309	0.061	0.9761	0.0153	secant used instead
Experimental dataset (our laboratory group)	Experimental Part III (-IPTG)	E	2-10 min	5	0.0285	0.0706	0.979	0.0152	secant used instead
Reference dataset (course-provided)	Reference Part II	A	2-6 min	3	0.136	0.0593	1	0.0016	yes
Reference dataset (course-provided)	Reference Part II	A	2-8 min	4	0.1086	0.1505	0.9595	0.0706	
Reference dataset (course-provided)	Reference Part II	A	2-10 min	5	0.1256	0.0825	0.967	0.0847	
Reference dataset (course-provided)	Reference Part II	B	2-6 min	3	0.1063	0.0423	0.9936	0.0241	
Reference dataset (course-provided)	Reference Part II	B	2-8 min	4	0.1059	0.0435	0.9974	0.0171	
Reference dataset (course-provided)	Reference Part II	B	2-10 min	5	0.1041	0.0509	0.9983	0.0155	yes
Reference dataset (course-provided)	Reference Part II	C	2-6 min	3	0.082	0.0023	0.9976	0.0114	yes
Reference dataset (course-provided)	Reference Part II	C	2-8 min	4	0.072	0.0355	0.9862	0.0269	
Reference dataset (course-provided)	Reference Part II	C	2-10 min	5	0.07	0.0435	0.9919	0.0232	
Reference dataset (course-provided)	Reference Part II	D	2-6 min	3	0.0243	0.0187	0.9899	0.0069	yes
Reference dataset (course-provided)	Reference Part II	D	2-8 min	4	0.0267	0.0105	0.9911	0.008	
Reference dataset (course-provided)	Reference Part II	D	2-10 min	5	0.0252	0.0167	0.9912	0.0086	
Reference dataset (course-provided)	Reference Part II	E	2-6 min	3	0.0238	0.0153	1	0.0004	
Reference dataset (course-provided)	Reference Part II	E	2-8 min	4	0.0226	0.019	0.9984	0.0029	
Reference dataset (course-provided)	Reference Part II	E	2-10 min	5	0.0229	0.018	0.9991	0.0025	yes
Reference dataset (course-provided)	Reference Part III (+IPTG)	A	2-6 min	3	0.097	0.0333	0.9358	0.0719	secant used instead
Reference dataset (course-provided)	Reference Part III (+IPTG)	A	2-8 min	4	0.0932	0.046	0.9701	0.0517	secant used instead
Reference dataset (course-provided)	Reference Part III (+IPTG)	A	2-10 min	5	0.092	0.0508	0.9843	0.0425	secant used instead
Reference dataset (course-provided)	Reference Part III (+IPTG)	B	2-6 min	3	0.0328	-0.0017	0.999	0.0029	yes
Reference dataset (course-provided)	Reference Part III (+IPTG)	B	2-8 min	4	0.0295	0.009	0.9918	0.0085	
Reference dataset (course-provided)	Reference Part III (+IPTG)	B	2-10 min	5	0.0285	0.0132	0.9942	0.0079	
Reference dataset (course-provided)	Reference Part III (+IPTG)	C	2-6 min	3	0.0208	0.0033	1	0.0004	yes
Reference dataset (course-provided)	Reference Part III (+IPTG)	C	2-8 min	4	0.0185	0.011	0.9897	0.0059	
Reference dataset (course-provided)	Reference Part III (+IPTG)	C	2-10 min	5	0.0176	0.0144	0.992	0.0058	
Reference dataset (course-provided)	Reference Part III (+IPTG)	D	2-6 min	3	0.0097	0.0103	0.9946	0.002	
Reference dataset (course-provided)	Reference Part III (+IPTG)	D	2-8 min	4	0.0092	0.012	0.9956	0.0019	
Reference dataset (course-provided)	Reference Part III (+IPTG)	D	2-10 min	5	0.0087	0.0142	0.9936	0.0026	yes
Reference dataset (course-provided)	Reference Part III (+IPTG)	E	2-6 min	3	0.0068	0.007	0.9959	0.0012	
Reference dataset (course-provided)	Reference Part III (+IPTG)	E	2-8 min	4	0.0066	0.0075	0.9979	0.0009	
Reference dataset (course-provided)	Reference Part III (+IPTG)	E	2-10 min	5	0.0062	0.0093	0.9935	0.0018	yes
Reference dataset (course-provided)	Reference Part III (+IPTG)	F	2-6 min	3	0.0035	0.0043	0.9932	0.0008	
Reference dataset (course-provided)	Reference Part III (+IPTG)	F	2-8 min	4	0.0033	0.005	0.9945	0.0008	
Reference dataset (course-provided)	Reference Part III (+IPTG)	F	2-10 min	5	0.0031	0.0058	0.9928	0.001	yes
Reference dataset (course-provided)	Reference Part III (-IPTG)	A	2-6 min	3	0.1332	0.0147	1	0.002	
Reference dataset (course-provided)	Reference Part III (-IPTG)	A	2-8 min	4	0.1352	0.008	0.9998	0.0054	
Reference dataset (course-provided)	Reference Part III (-IPTG)	A	2-10 min	5	0.132	0.021	0.9993	0.0126	yes
Reference dataset (course-provided)	Reference Part III (-IPTG)	B	2-6 min	3	0.0785	0.053	0.9923	0.0196	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	B	2-8 min	4	0.0706	0.0795	0.9878	0.0248	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	B	2-10 min	5	0.0643	0.1043	0.9836	0.0304	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	C	2-6 min	3	0.0502	0.0627	0.985	0.0176	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	C	2-8 min	4	0.0443	0.0825	0.9805	0.0198	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	C	2-10 min	5	0.0386	0.1055	0.9658	0.0265	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	D	2-6 min	3	0.0253	0.0633	0.983	0.0094	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	D	2-8 min	4	0.0214	0.076	0.9704	0.0118	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	D	2-10 min	5	0.0225	0.0718	0.9842	0.0104	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	E	2-6 min	3	0.0125	0.0933	0.9745	0.0057	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	E	2-8 min	4	0.0092	0.1045	0.9018	0.0095	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	E	2-10 min	5	0.0076	0.1109	0.889	0.0097	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	F	2-6 min	3	0.0075	0.0337	0.9643	0.0041	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	F	2-8 min	4	0.006	0.0385	0.9425	0.0047	secant used instead
Reference dataset (course-provided)	Reference Part III (-IPTG)	F	2-10 min	5	0.0051	0.0421	0.9322	0.0051	secant used instead

Table A7. Regression over every candidate initial window (2 to 6, 2 to 8, 2 to 10 min) for every curve; “Selected” marks the window used, or notes that the secant replaced it.

Role	Dataset	Tube	increments	secant 0-2 (origin)	secant 2-4	secant 2-6	regression 2-6	regression initial window	initial window	regression all 5 points	rule (V0 used)	rule estimator	secant 0-2 offset-corrected
experimental	Experimental Part II	A	0.268 / 0.273 / 0.320 / 0.191	0.138	0.134	0.1352	0.1353	0.1428	2-8 min	0.1348	0.1428	regression 2-8 min	0.1316

Role	Dataset	Tube	increments	secant 0-2 (origin)	secant 2-4	secant 2-6	regression 2-6	regression initial window	initial window	regression all 5 points	rule (V0 used)	rule estimator	secant 0-2 offset-corrected
experimental	Experimental Part II	B	0.174 / 0.179 / 0.161 / 0.166	0.09	0.087	0.0883	0.0883	0.085	2-10 min	0.085	0.085	regression 2-10 min	0.0836
experimental	Experimental Part II	C	0.122 / 0.120 / 0.124 / 0.084	0.0665	0.061	0.0605	0.0605	0.0609	2-8 min	0.0572	0.0609	regression 2-8 min	0.0601
experimental	Experimental Part II	D	0.056 / 0.055 / 0.059 / 0.048	0.0325	0.028	0.0277	0.0277	0.0275	2-10 min	0.0275	0.0275	regression 2-10 min	0.0261
experimental	Experimental Part II	E	0.043 / 0.037 / 0.042 / 0.036	0.0235	0.0215	0.02	0.02	0.0197	2-10 min	0.0197	0.0197	regression 2-10 min	0.0171
experimental	Experimental Part III (+IPTG)	A	0.157 / 0.158 / 0.150 / 0.152	0.0815	0.0785	0.0787	0.0788	0.0771	2-10 min	0.0771	0.0771	regression 2-10 min	0.0751
experimental	Experimental Part III (+IPTG)	B	0.237 / 0.242 / 0.247 / 0.111	0.1345	0.1185	0.1197	0.1198	0.121	2-8 min	0.1082	0.121	regression 2-8 min	0.1281
experimental	Experimental Part III (+IPTG)	C	0.078 / 0.086 / 0.060 / 0.058	0.04	0.039	0.041	0.041	0.041	2-6 min	0.0355	0.041	regression 2-6 min	0.0336
experimental	Experimental Part III (+IPTG)	D	0.041 / 0.034 / 0.035 / 0.037	0.0205	0.0205	0.0188	0.0187	0.0181	2-10 min	0.0181	0.0181	regression 2-10 min	0.0141
experimental	Experimental Part III (-IPTG)	A	0.221 / 0.216 / 0.242 / 0.123	0.1295	0.1105	0.1092	0.1092	0.1126	2-8 min	0.1031	0.1126	regression 2-8 min	0.1231
experimental	Experimental Part III (-IPTG)	B	0.212 / 0.219 / 0.209 / 0.201	0.112	0.106	0.1078	0.1077	0.1055	2-10 min	0.1055	0.1055	regression 2-10 min	0.1056
experimental	Experimental Part III (-IPTG)	C	0.188 / 0.161 / 0.193 / 0.138	0.1	0.094	0.0873	0.0873	0.0894	2-8 min	0.0857	0.0894	regression 2-8 min	0.0936
experimental	Experimental Part III (-IPTG)	D	0.184 / 0.165 / 0.123 / 0.090	0.101	0.092	0.0873	0.0872	0.0872	2-6 min	0.0706	0.0872	regression 2-6 min	0.0946
experimental	Experimental Part III (-IPTG)	E	0.090 / 0.048 / 0.052 / 0.045	0.0555	0.045	0.0345	0.0345	0.0345	2-6 min	0.0285	0.0555	0-2 min secant (no initial linear portion)	0.0491
reference	Reference Part II	A	0.270 / 0.274 / 0.089 / 0.442	0.166	0.135	0.136	0.136	0.136	2-6 min	0.1256	0.136	regression 2-6 min	0.1596
reference	Reference Part II	B	0.242 / 0.183 / 0.220 / 0.194	0.1225	0.121	0.1063	0.1063	0.1041	2-10 min	0.1041	0.1041	regression 2-10 min	0.1161
reference	Reference Part II	C	0.150 / 0.178 / 0.093 / 0.144	0.0855	0.075	0.082	0.082	0.082	2-6 min	0.07	0.082	regression 2-6 min	0.0791
reference	Reference Part II	D	0.040 / 0.057 / 0.062 / 0.033	0.035	0.02	0.0243	0.0243	0.0243	2-6 min	0.0252	0.0243	regression 2-6 min	0.0286
reference	Reference Part II	E	0.047 / 0.048 / 0.040 / 0.050	0.0315	0.0235	0.0238	0.0238	0.0229	2-10 min	0.0229	0.0229	regression 2-10 min	0.0251
reference	Reference Part III (+IPTG)	A	0.282 / 0.106 / 0.198 / 0.182	0.099	0.141	0.097	0.097	0.092	2-10 min	0.092	0.099	0-2 min secant (no initial linear portion)	0.0926
reference	Reference Part III (+IPTG)	B	0.062 / 0.069 / 0.043 / 0.055	0.0325	0.031	0.0328	0.0328	0.0328	2-6 min	0.0285	0.0328	regression 2-6 min	0.0261
reference	Reference Part III (+IPTG)	C	0.041 / 0.042 / 0.026 / 0.033	0.0225	0.0205	0.0208	0.0208	0.0208	2-6 min	0.0176	0.0208	regression 2-6 min	0.0161
reference	Reference Part III (+IPTG)	D	0.022 / 0.017 / 0.017 / 0.014	0.0145	0.011	0.0098	0.0097	0.0087	2-10 min	0.0087	0.0087	regression 2-10 min	0.0081
reference	Reference Part III (+IPTG)	E	0.015 / 0.012 / 0.013 / 0.009	0.01	0.0075	0.0068	0.0068	0.0062	2-10 min	0.0062	0.0062	regression 2-10 min	0.0036
reference	Reference Part III (+IPTG)	F	0.008 / 0.006 / 0.006 / 0.005	0.0055	0.004	0.0035	0.0035	0.0031	2-10 min	0.0031	0.0031	regression 2-10 min	-0.0009
reference	Reference Part III (-IPTG)	A	0.264 / 0.269 / 0.279 / 0.234	0.141	0.132	0.1332	0.1332	0.132	2-10 min	0.132	0.132	regression 2-10 min	0.1346
reference	Reference Part III (-IPTG)	B	0.181 / 0.133 / 0.112 / 0.095	0.101	0.0905	0.0785	0.0785	0.0785	2-6 min	0.0643	0.101	0-2 min secant (no initial linear portion)	0.0946
reference	Reference Part III (-IPTG)	C	0.122 / 0.079 / 0.068 / 0.043	0.078	0.061	0.0502	0.0502	0.0502	2-6 min	0.0386	0.078	0-2 min secant (no initial linear portion)	0.0716
reference	Reference Part III (-IPTG)	D	0.062 / 0.039 / 0.029 / 0.061	0.055	0.031	0.0252	0.0253	0.0253	2-6 min	0.0225	0.055	0-2 min secant (no initial linear portion)	0.0486
reference	Reference Part III (-IPTG)	E	0.032 / 0.018 / 0.005 / 0.009	0.058	0.016	0.0125	0.0125	0.0125	2-6 min	0.0076	0.058	0-2 min secant (no initial linear portion)	0.0516
reference	Reference Part III (-IPTG)	F	0.020 / 0.010 / 0.007 / 0.006	0.0235	0.01	0.0075	0.0075	0.0051	2-10 min	0.0051	0.0235	0-2 min secant (no initial linear portion)	0.0171

Table A7b. Two-point secants, window regressions and the rule's V_0 for every curve (ΔA_{415} min^{-1}). For linear curves the 2 to 6 min secant and the 2 to 6 min regression agree within rounding, as they must for three equally spaced points; the 0 to 2 min secant exceeds them wherever the intercept is positive.

3.3 Units

V_0 (mM min^{-1} in the cuvette) = slope/ε. The stopped sample is 2 mL reaction + 1 mL Na_2CO_3, so the reaction-mixture rate is 1.5 × the cuvette rate; multiplying by 12.0 mL gives µmol ONP min^{-1} per assay. With ε = 5.22 A mM^{-1} cm^{-1}, 1 ΔA_{415} min^{-1} = 3.447 µmol min^{-1} per assay (1/5.2224 × 1.5 × 12.0 = 3.4467; dimensionally A min^{-1} ÷ (A mM^{-1}) × mL = µmol min^{-1}). The manual's abbreviated instruction, "divide the slope by ε", yields the cuvette rate in mM min^{-1}; the two volume factors are needed to express an amount per minute. K_m and K_i do not depend on the factor (it cancels in slope/intercept ratios); V_max scales with it, and per mL of enzyme the per-assay value is multiplied by 5 (0.2 mL enzyme per assay).

3.4 Calibration

ε is the slope of the least-squares line through the five standards from 0.083 to 0.433 mM (free intercept -0.0224 A, p = 0.50, so not distinguishable from zero). Alternatives are tabulated below. The 0.533 mM standard read 3.236 A, 17% above the line; its transmittance is 0.06% of the incident light. The instrument's specified photometric linear range extends to 3.5 A (at 260 nm), so the reading is within the nominal range but close to its limit, where small stray-light or detector-linearity errors are amplified; stray light would depress a reading, whereas this one is high, so a pipetting difference in the 1.6 mL standard is as plausible as an instrumental cause. The data cannot decide, which is why the point is excluded on the operational criterion (increment rule) and its effect quantified rather than explained. Matrix: the Part I standards and blank contain Na_2CO_3 as the stopped samples do, but the stopped samples contain ONPG (± IPTG) and enzyme while the Part II/III blank is Z buffer alone; ONPG does not absorb appreciably at 415 nm, and the small positive intercepts of the linear progress curves (Table A5) are the empirical measure of any residual mismatch.

Calibration	Points	Highest [ONP] (mM)	Highest A415	Epsilon (A per mM per cm)	SE	Intercept (A)	R2	Vmax relative to primary
five standards, free intercept (used)	5	0.433	2.256	5.222	0.104	-0.0224	0.9988	1
five standards, through the origin	5	0.433	2.256	5.151	0.043	0	0.9986	1.014
zero-concentration point + five standards, free intercept	6	0.433	2.256	5.186	0.072	-0.0109	0.9992	1.007
all six nonzero standards, free intercept	6	0.533	3.236	6.008	0.426	-0.1792	0.9802	0.869
zero point + all six standards, free intercept	7	0.533	3.236	5.796	0.332	-0.0992	0.9839	0.901

Table A8. Calibration variants and their effect on V_max (K_m and K_i unchanged).

3.5 Kinetic models

Lineweaver–Burk by ordinary least squares (course method); unweighted nonlinear least squares of V_0 = V_max[S]/(K_m + [S]) on the velocity scale with the bounded "port" algorithm from a grid of starting values (a fit that runs to the K_m bound of 100 mM is reported as "no plateau within the tested range" rather than as a number); K_i = [I]/(K_m,app/K_m − 1) as in the manual; and three global models fitted to both series simultaneously: competitive, V = V_max[S]/(K_m(1 + [I]/K_i) + [S]); pure noncompetitive, V = V_max[S]/((K_m + [S])(1 + [I]/K_i)); mixed, V = V_max[S]/(K_m(1 + [I]/K_i) + S). Competitive and noncompetitive have three parameters each, so their residual sums of squares compare directly; the mixed model nests both. No variance weights were used.

4. Primary results (reference dataset)

Main figures (`figures/`) are drawn from the reference dataset (course-provided measurements, used for the Results section at the instructor's direction); supplementary figures (`figures/supplementary/`) hold the experimental dataset (our own printouts), diagnostics and sensitivity analyses. All figures: 600 dpi PNG plus vector PDF, generated by `R/07_generate_figures.R` and `R/07b_supplementary_figures.R`. Conventions: filled symbol, observation used for V_0; open symbol, observation retained but not used; solid line, fitted initial window; dashed segment from the origin, 0–2 min secant through the assumed origin (no reading exists at t = 0); dotted line, extrapolation; ↑ marks a lower-bound V_0. Each condition was measured once, so no error bars are shown.

Figure 1. o-Nitrophenol absorbs maximally at 415 nm and obeys the Beer–Lambert law to 0.43 mM

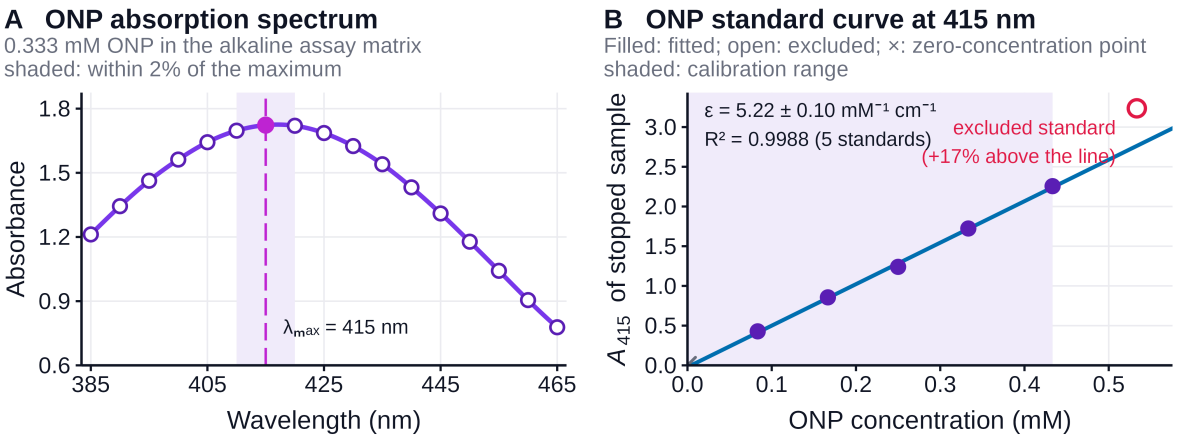


Figure 1

(A) Absorbance spectrum of 0.333 mM ONP (manual tube 4: 1.0 mL of 1 mM ONP + 1.0 mL Z buffer + 1.0 mL 1 M Na_2CO_3) read from 465 to 385 nm in 5 nm steps against the Z buffer/Na_2CO_3 blank (reference Part I). Points are the readings, the line a spline through them, the dashed line the maximum at 415 nm (A = 1.723); the shaded band (410–420 nm) lies within 2% of the maximum, so 415 nm was used for all kinetic measurements. **(B)** Standard curve at 415 nm for 1 mM ONP diluted to 0.083–0.533 mM in the same 3.0 mL matrix ([ONP] = 1 mM × V/3.0 mL; mapping of the reported readings to the manual's preparations in Table S1). Filled points and the line are the least-squares calibration over the linear region (shaded, 0.083–0.433 mM, chosen by the same increment rule used for progress curves): A_{415} = 5.222[ONP] − 0.022, R^2 = 0.9988, n = 5, giving ε = 5.22 ± 0.10 A mM^{-1} cm^{-1} (5222 M^{-1} cm^{-1}). The zero-concentration point (×, reported as 0) is shown but not fitted. The 0.533 mM standard (open, A = 3.24) lay +17% above the line and was excluded; the cause of the deviation cannot be identified from a single reading (see text). The calibration converts every velocity in this report from ΔA_{415} min^{-1} to product formed.

Figure 2. Initial velocity rose approximately in proportion to the amount of β-galactosidase

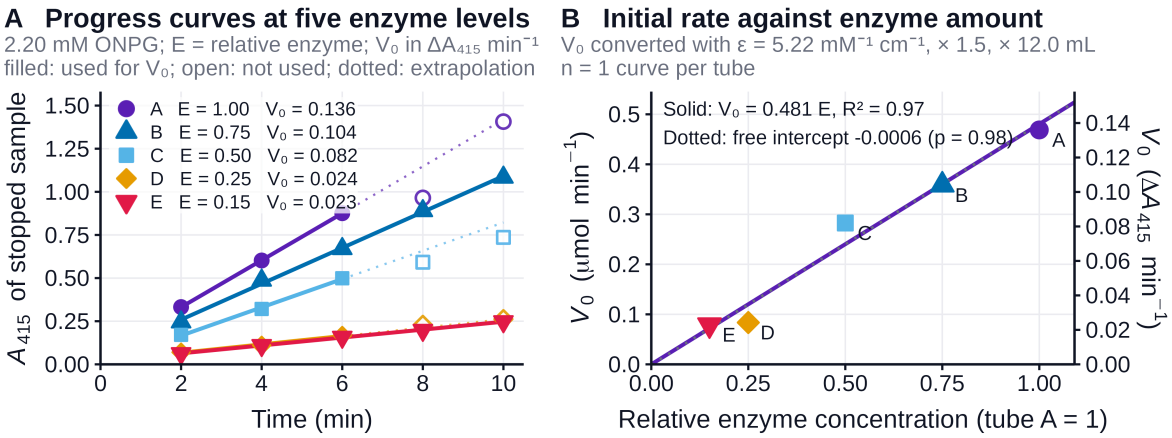


Figure 2

(A) Progress curves (A_{415} of 2.0 mL aliquots stopped with 1.0 mL 1 M Na_2CO_3) for 2.20 mM ONPG hydrolysed by 1.00, 0.75, 0.50, 0.25 and 0.15 relative amounts of the standardised enzyme (tubes A–E; reference Part II). Lines are least-squares fits to the initial linear window chosen by the operational rule (Methods); the 8-min readings of tubes A and C fell far below their initial trend and were excluded (open), while tubes B and E were linear to 10 min. V_0 (legend, ΔA_{415} min^{-1}) ranged from 0.136 (A) to 0.023 (E), a 5.9-fold span for a 6.7-fold change in enzyme. **(B)** V_0, converted with ε = 5.22 mM^{-1} cm^{-1} to µmol ONP min^{-1} per 12.0 mL assay (right axis in ΔA_{415} min^{-1}), against relative enzyme concentration. Solid line: proportional model through the origin, V_0 = 0.481 × E (R^2 about the mean 0.972); dotted: free-intercept fit, intercept -0.0006 µmol min^{-1}, not distinguishable from zero (p = 0.98). Relative rates were 1.00, 0.77, 0.60, 0.18, 0.17 for relative enzyme 1.00, 0.75, 0.50, 0.25, 0.15: velocity rose approximately in proportion to enzyme, but the rate per unit enzyme ranged from 0.71 (D) to 1.21 (C) and tubes D and E gave nearly equal rates despite a 1.7-fold difference in enzyme.

Figure 3. β-Galactosidase approached saturation with ONPG

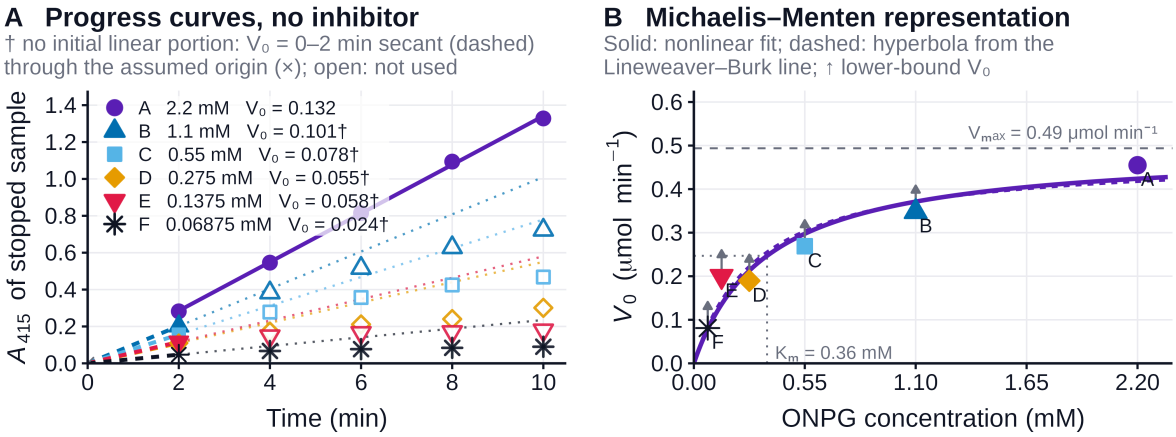


Figure 3

(A) Progress curves for 0.2 mL enzyme acting on 2.20, 1.10, 0.55, 0.275, 0.1375 and 0.069 mM ONPG in 12.0 mL Z buffer at room temperature (tubes A–F; reference Part III, no inhibitor). Only tube A stayed linear to 10 min (V_0 = 0.132 ΔA_{415} min^{-1}, regression); in tubes B–F the increment fell from the first interval onwards (†), so their V_0 is the 0–2 min secant through the assumed origin (dashed, × at the origin; dotted extrapolation). The secant is a lower bound on the initial rate if the absorbance offset at t = 0 is zero; the intercepts of the curves that were linear (median 0.013 A, range -0.020 to 0.059 A) bound the possible overstatement at about 0.006 ΔA_{415} min^{-1}. Later points (open) were not used. **(B)** V_0 in µmol ONP min^{-1} per assay against [ONPG]. Course-required Lineweaver–Burk analysis (dashed hyperbola): K_m = 0.31 mM, V_{max} = 0.477 µmol min^{-1} (R^2 = 0.928). Direct nonlinear least-squares fit on the velocity scale (solid): K_m = 0.36 ± 0.13 mM, V_{max} = 0.494 ± 0.059 µmol min^{-1} (R^2 = 0.919); dotted guides mark K_m and $V_{max}/2$. Velocity rose 5.6-fold from 0.069 to 2.20 mM and approached a plateau; because five of six V_0 values are lower bounds, K_m is an upper-range estimate under the tested conditions.

Figure 4. IPTG raised the apparent K_m of β-galactosidase for ONPG

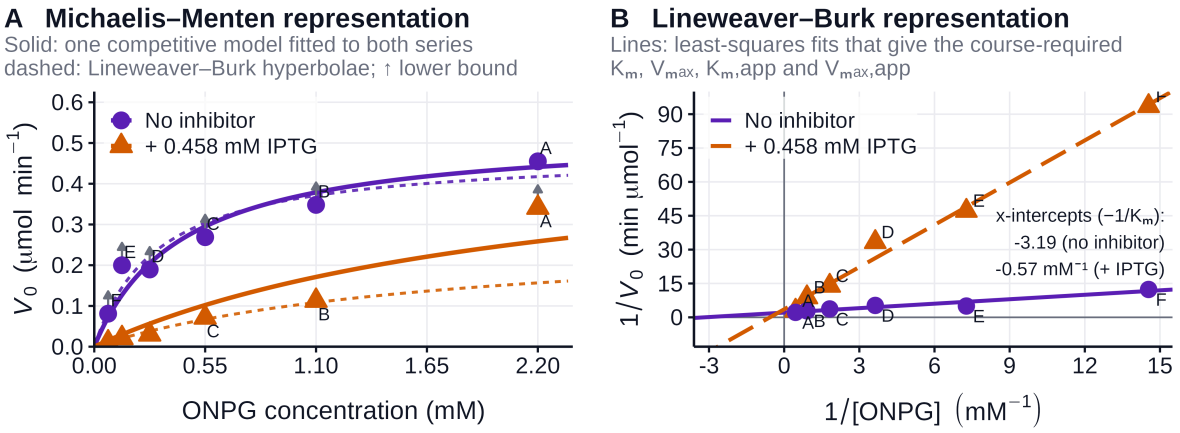


Figure 4

(A) V_0 against [ONPG] without inhibitor (blue circles, from Figure 3) and with 0.458 mM IPTG (magenta triangles; 1.0 mL of 5.5 mM IPTG per 12.0 mL tube; reference Part III, tubes A–F). Dashed curves are the hyperbolae implied by the two Lineweaver–Burk lines in B; solid curves are a single competitive-inhibition model fitted to both series at once (shared V_{max} = 0.53 ± 0.07 µmol min^{-1}, K_m = 0.44 ± 0.17 mM, K_i = 0.107 ± 0.042 mM; the shared V_{max} is an assumption of this model, not a result). With IPTG, V_0 fell to 75% of the uninhibited value at 2.20 mM ONPG and to 13% at 0.069 mM: inhibition grew as substrate fell. **(B)** Double-reciprocal plot of the same velocities. No inhibitor: $1/V_0$ = 0.66(1/[S]) + 2.10, R^2 = 0.928, giving V_{max} = 0.477 µmol min^{-1} and K_m = 0.31 mM. + IPTG: $1/V_0$ = 6.24(1/[S]) + 3.55, R^2 = 0.988, giving apparent V_{max} = 0.282 µmol min^{-1} and apparent K_m = 1.76 mM (5.6-fold increase). K_i = [I]/($K_{m,app}/K_m$ − 1) = 0.458/(1.76/0.31 − 1) = 0.100 mM. The inhibited series shows no plateau within the tested range, so its apparent V_{max} rests on an extrapolated intercept (3.55 ± 2.33) and is poorly determined; the shift of the x-intercept (−1/K_m) from -3.19 to -0.57 mM^{-1} is the better-determined feature. A pure noncompetitive model fitted to the same twelve points (shared K_m, V_{max} reduced) left a residual sum of squares 1.9 times larger than the competitive model (0.0371 against 0.0194); the pattern is therefore consistent with competitive inhibition, with K_i of the order of 0.1 mM.

Supplementary figures (`figures/supplementary/`)

Figure S1. Progress curves of the experimental dataset beside the reference dataset

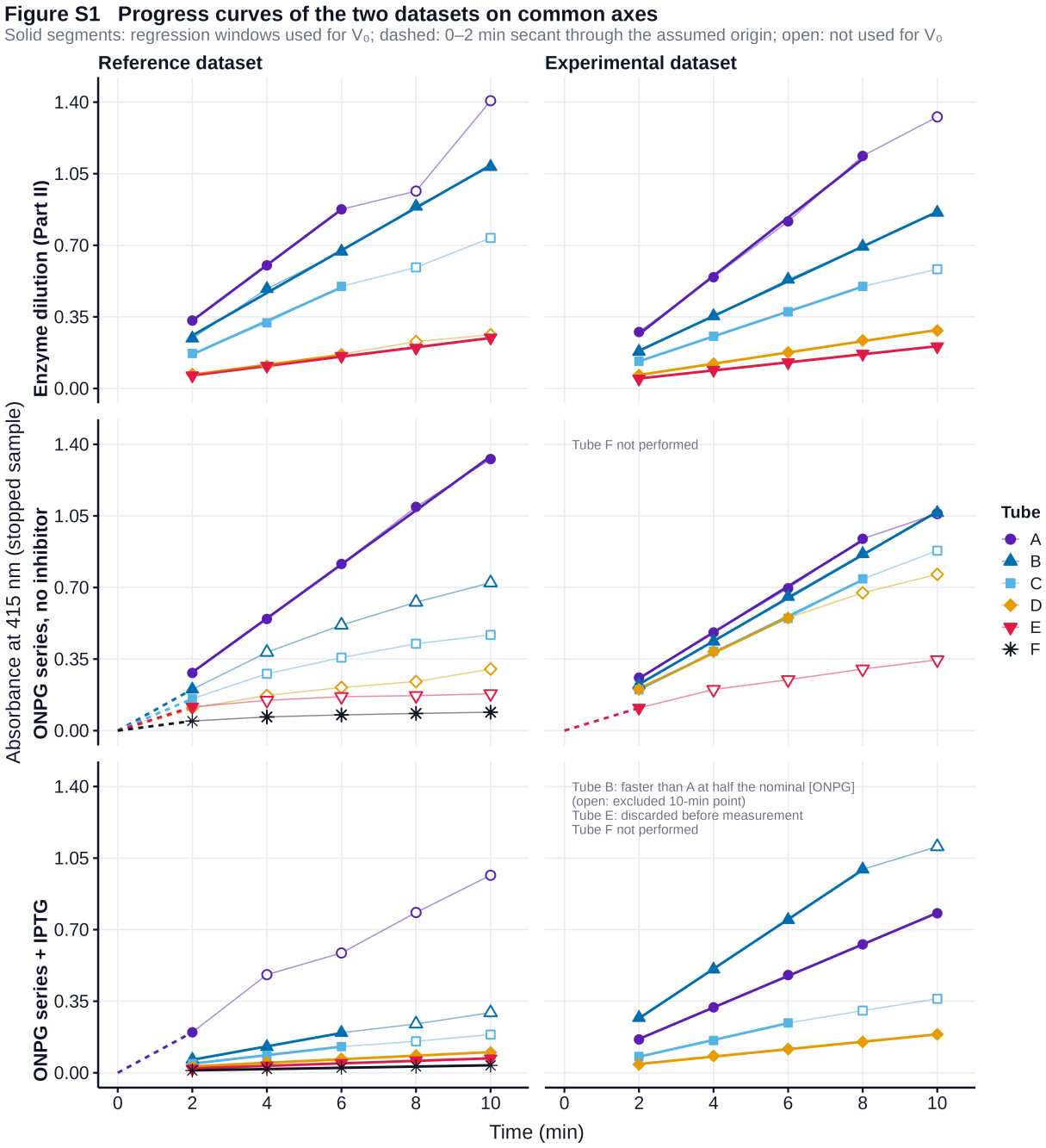


Figure S1

Progress curves from our 14 GENESYS 10S printouts (right column; 7 and 11 September 2026) plotted on the same axes as the reference dataset (left column) for the three experiments. Solid segments are the regression windows and dashed segments the 0–2 min secants used for V_0 under the identical rule. The standard tube (A: 0.2 mL enzyme, 2.20 mM ONPG) agreed between the two datasets within 5% (enzyme dilution) and 15% (substrate series). Our low-substrate tubes without inhibitor stayed linear longer than the reference tubes (only tube E was non-linear from the outset), and our inhibited tubes C and D ran about twice as fast as the reference tubes. Our inhibited tube B (open) ran faster than tube A at half the nominal substrate and is excluded from kinetic fits; our inhibited tube E has no data (sample discarded); tube F was not performed on either day.

Figure S2. Increment diagnostics behind every V_0 decision

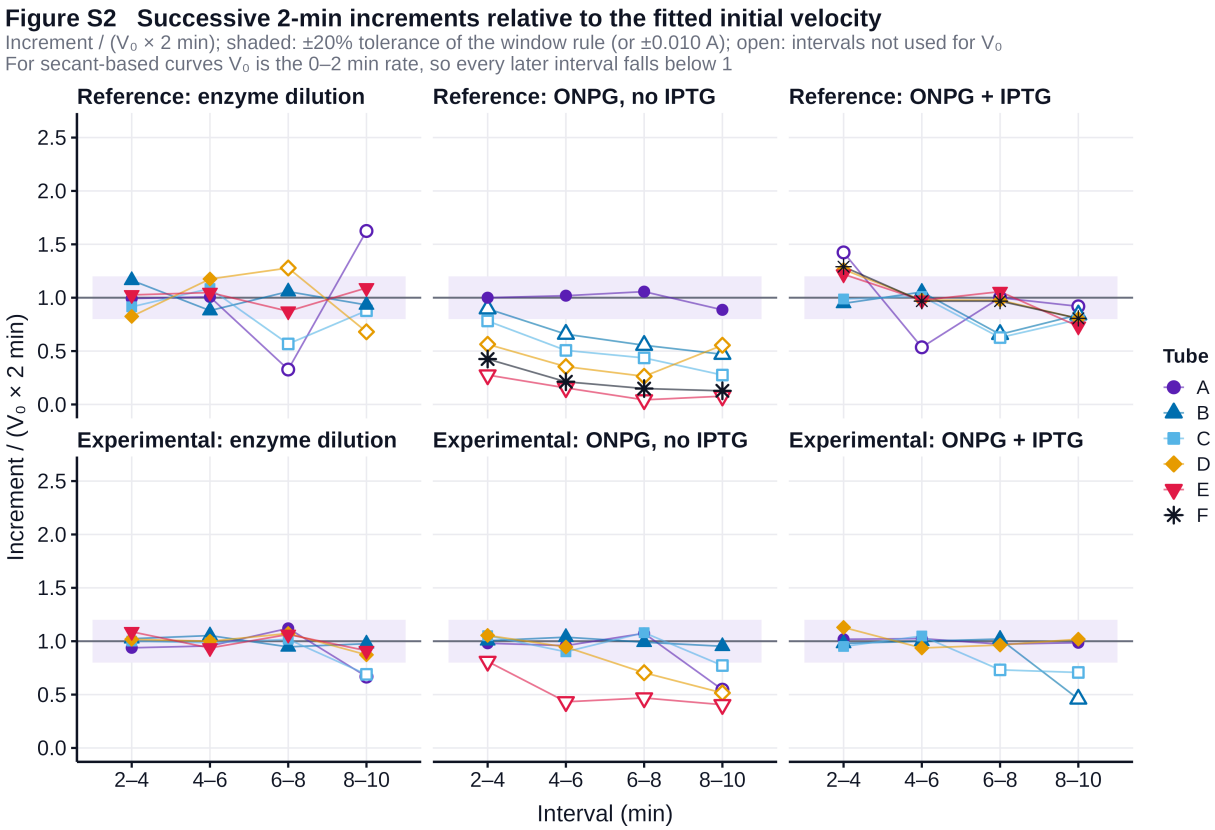


Figure S2

For each of the 31 measured curves, every successive 2-min increment divided by the increment predicted by the fitted V_0 ($V_0 \times 2$ min). A value of 1 means the interval ran at the fitted initial rate; the shaded band is the acceptance window of the rule (±20% or ±0.010 A, whichever is larger). Open symbols are intervals not used for V_0. For curves with no initial linear portion (reference no-IPTG B–F, reference +IPTG A, experimental no-IPTG E) the reference V_0 is the 0–2 min secant, so all later intervals fall below 1, showing how quickly the rate declined: to 35–60% of the initial rate by the second interval in the reference uninhibited tubes B–F.

Figure S3. Sensitivity of the kinetic parameters to the V_0 estimator

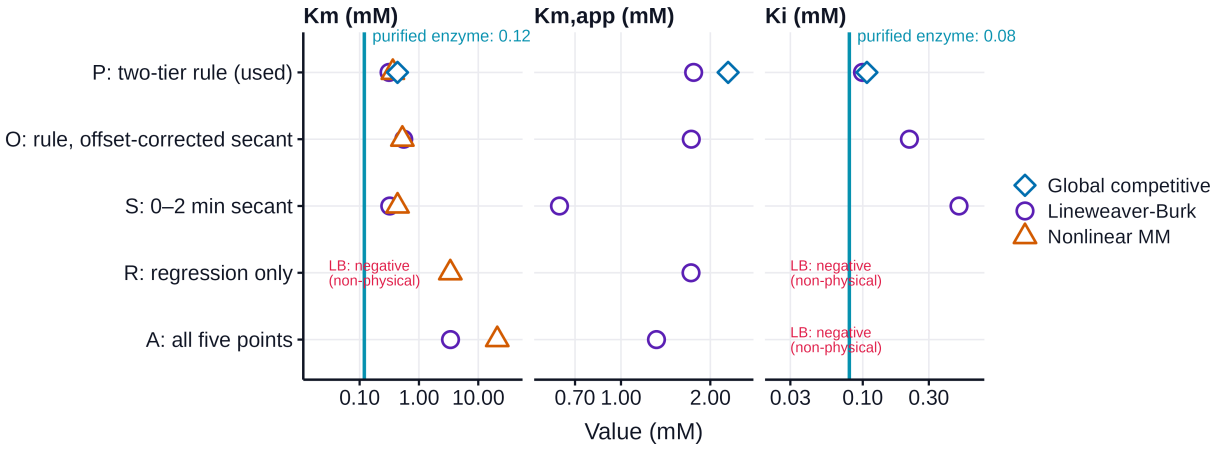


Figure S3

K_m (no inhibitor), apparent K_m (+IPTG) and K_i from the reference dataset under five ways of estimating V_0: A, regression on all five points of every curve; R, the initial-window regression for every curve; S, the 0–2 min secant for every curve; O, the two-tier rule with each secant corrected for the

median t = 0 offset of the linear curves (0.013 A); P, the two-tier rule used in the Results. Circles, Lineweaver–Burk; triangles, nonlinear Michaelis–Menten; diamond, global competitive fit. With regression-only V_0 the uninhibited Lineweaver–Burk line has a negative intercept (non-physical K_m and V_{max}, noted in red), and with all five points K_m exceeds 3 mM, because both estimators average over the falling phase of the curved low-substrate tubes. The offset correction raises K_m from 0.31 to 0.55 mM and K_i from 0.10 to 0.22 mM; the uniform secant (S) gives K_i = 0.50 mM because it is biased upward for the inhibited tubes. K_i is therefore known only to within a factor of two to five. Cyan lines: K_m = 0.12 mM and K_i = 0.08 mM reported for purified E. coli β-galactosidase with ONPG in TES buffer, pH 7.0, 25 °C.

Figure S4. Kinetic analysis of the experimental dataset

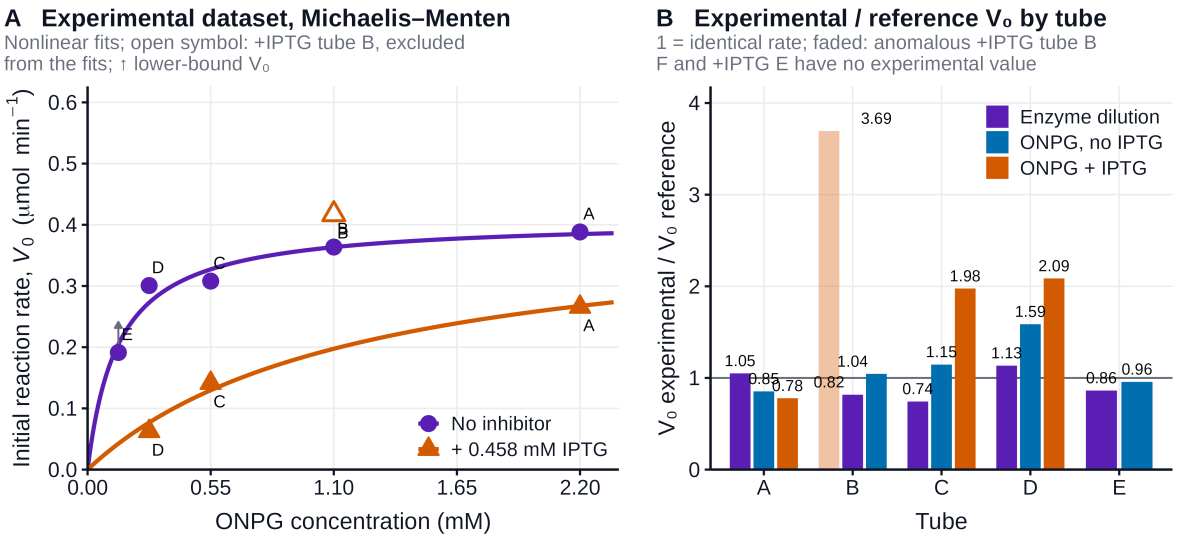


Figure S4

(A) Michaelis–Menten representation of our Part III data with the same V_0 rule and units as the Results, with the anomalous +IPTG tube B (open) excluded and +IPTG tube E absent. Nonlinear fits: K_m = 0.14 mM, V_{max} = 0.41 µmol min⁻¹ without inhibitor; apparent K_m = 1.21 mM, apparent V_{max} = 0.41 with IPTG (ratio 1.01); K_i = 0.059 mM. **(B)** Ratio of our V_0 to the reference V_0 for every tube measured in both datasets. The standard tube agreed within 5–15%; our uninhibited tube D and inhibited tubes C and D ran 1.6- to 2.1-fold faster; our inhibited tube B (faded) ran 3.7-fold faster than the reference tube B.

4.1 Calibration

Reported tube	Manual preparation	ONP 1 mM (mL)	[ONP] in cuvette (mM)	A415	A415 / [ONP] (per mM)	Fitted
1	Blank	0	0	0	NA	no (zero point)
2	Standard 1	0.25	0.0833	0.428	5.14	yes
3	Standard 2	0.5	0.1667	0.856	5.14	yes
4	Standard 3	0.75	0.25	1.24	4.96	yes
5	Standard 4	1	0.3333	1.723	5.17	yes
6	Standard 5	1.3	0.4333	2.256	5.21	yes
7	Standard 6	1.6	0.5333	3.236	6.07	no (+17%)

Table A9. ONP standard curve. λ_max = 415 nm (A = 1.723; 410 to 420 nm within 2% of the maximum). Calibration over 5 standards: A_{415} = 5.222[ONP] − 0.022, R^2 = 0.9988, ε = 5.22 ± 0.10 A mM⁻¹ cm⁻¹ (5222 M⁻¹ cm⁻¹). For comparison, ε ≈ 4.5 mM⁻¹ cm⁻¹ at 420 nm is the value used in Miller's classical assay in the same alkaline stop; at pH 7.0, where ONP is largely protonated, Juers et al. used 2.65 mM⁻¹ cm⁻¹ at 420 nm, which is not comparable with the alkaline value measured here.

4.2 Enzyme dilution

Tube	Relative enzyme	V0 (dA415/min)	V0 (umol/min)	Relative rate	Rate / enzyme	Points used	V0 experimental (dA415/min)	Relative rate, experimental
A	1	0.136	0.469	1	1	2-6 min	0.1428	1
B	0.75	0.1041	0.359	0.77	1.02	2-10 min	0.085	0.6
C	0.5	0.082	0.283	0.6	1.21	2-6 min	0.0609	0.43
D	0.25	0.0243	0.084	0.18	0.71	2-6 min	0.0275	0.19
E	0.15	0.0229	0.079	0.17	1.12	2-10 min	0.0197	0.14

Table A10. Enzyme dilution, reference (primary) with the experimental values alongside. Reference: proportional slope 0.481 µmol min⁻¹ per unit relative enzyme, R^2 0.972; free intercept -0.0006 µmol min⁻¹, p = 0.98; rate per unit enzyme 0.71 (D) to 1.21 (C). Experimental: R^2 0.963, relative rates 1.00, 0.60, 0.43, 0.19, 0.14. In both datasets the rate per unit enzyme is lowest for tube D (0.25); in the reference dataset tubes D and E gave nearly the same rate (relative rates 0.18 and 0.17 for enzyme 0.25 and 0.15). The trend is approximately, not exactly, proportional.

4.3 Substrate series and inhibition

Series	Tube	[S] (mM)	2 min	4 min	6 min	8 min	10 min	V0 (dA/min)	V0 (umol/min)	1/[S]	1/V0
no inhibitor	A	2.2	0.282	0.546	0.815	1.094	1.328	0.132	0.455	0.45	2.2
no inhibitor	B	1.1	0.202	0.383	0.516	0.628	0.723	0.101†	0.348†	0.91	2.87
no inhibitor	C	0.55	0.156	0.278	0.357	0.425	0.468	0.078†	0.269†	1.82	3.72
no inhibitor	D	0.275	0.11	0.172	0.211	0.24	0.301	0.055†	0.190†	3.64	5.28
no inhibitor	E	0.1375	0.116	0.148	0.166	0.171	0.18	0.058†	0.200†	7.27	5
no inhibitor	F	0.06875	0.047	0.067	0.077	0.084	0.09	0.0235†	0.081†	14.55	12.35
+ 0.458 mM IPTG	A	2.2	0.198	0.48	0.586	0.784	0.966	0.099†	0.341†	0.45	2.93
+ 0.458 mM IPTG	B	1.1	0.065	0.127	0.196	0.239	0.294	0.03275	0.113	0.91	8.86
+ 0.458 mM IPTG	C	0.55	0.045	0.086	0.128	0.154	0.187	0.02075	0.072	1.82	13.98
+ 0.458 mM IPTG	D	0.275	0.029	0.051	0.068	0.085	0.099	0.0087	0.030	3.64	33.35
+ 0.458 mM IPTG	E	0.1375	0.02	0.035	0.047	0.06	0.069	0.00615	0.021	7.27	47.18
+ 0.458 mM IPTG	F	0.06875	0.011	0.019	0.025	0.031	0.036	0.0031	0.011	14.55	93.59

Table A11. Data-sheet view of the primary substrate series: absorbances, V_0 and reciprocals.

Condition	Method	Km	Vmax	Ki	Fit
No inhibitor	Lineweaver-Burk	0.314	0.477	NA	R2 0.928
No inhibitor	Nonlinear MM	0.363 ± 0.128	0.494 ± 0.059	NA	R2 0.919
+ 0.458 mM IPTG	Lineweaver-Burk	1.757 (app.)	0.282 (app.)	0.100	R2 0.988
+ 0.458 mM IPTG	Nonlinear MM	n.d.	n.d.	NA	R2 0.962
Both series	Global, competitive	0.435 ± 0.166	0.529 ± 0.072	0.107 ± 0.042	RSS 0.0194
Both series	Global, noncompetitive	0.744 ± 0.382	0.626 ± 0.139	0.433 ± 0.181	RSS 0.0371

Method	Km (mM)	Km,app (mM)	Km,app/Km	Vmax (umol/min)	Vmax,app (umol/min)	Vmax,app/Vmax	[IPTG] (mM)	Ki, competitive (mM)	Ki if noncompetitive (mM)
Lineweaver-Burk	0.314	1.757	5.6	0.477	0.282	0.59	0.458	0.1	0.662
Nonlinear MM	0.363	NA	NA	0.494	NA	NA	0.458	NA	NA
Global nonlinear competitive	0.435	2.297	5.28	0.529	0.529	1	0.458	0.107	NA

Table A12. Primary kinetic parameters and inhibition analysis. The +IPTG Lineweaver–Burk intercept (3.55 ± 2.33) is poorly determined because the inhibited velocities halve with each halving of substrate from tube B downward (first-order behaviour) and only tube A hints at a plateau; the apparent V_max (0.282 µmol min⁻¹) therefore rests on extrapolation, and the nonlinear fit of the inhibited series alone runs to the K_m bound. A test of the intercept against zero says nothing about whether the two series share a V_max; the question is addressed by the model comparison in Table A13. The K_i that a noncompetitive interpretation would assign to the apparent V_max change (0.66 mM) is listed for completeness.

Model	Parameter	Estimate	SE	RSS	AIC	n	Converged	Note
global competitive	Vmax	0.5285	0.072	0.01941	-35.07	12	TRUE	
global competitive	Km	0.4351	0.1662	0.01941	-35.07	12	TRUE	
global competitive	Ki	0.1071	0.0418	0.01941	-35.07	12	TRUE	
global noncompetitive	Vmax	0.6262	0.1388	0.03713	-27.28	12	TRUE	
global noncompetitive	Km	0.7438	0.3824	0.03713	-27.28	12	TRUE	
global noncompetitive	Ki	0.4333	0.1806	0.03713	-27.28	12	TRUE	
global mixed	Vmax	0.5286	0.0778	0.01942	-33.06	12	TRUE	
global mixed	Km	0.4352	0.1788	0.01942	-33.06	12	TRUE	
global mixed	Ki	0.1072	0.0792	0.01942	-33.06	12	TRUE	
global mixed	Kip	500	422500	0.01942	-33.06	12	TRUE	at upper bound (model reduces to competitive)

Table A13. Global fits to both series (12 points). Competitive: V_max = 0.529 ± 0.072 µmol min⁻¹, K_m = 0.435 ± 0.166 mM, K_i = 0.107 ± 0.042 mM, RSS 0.0194, AIC -35.1. Pure noncompetitive (same number of parameters): K_i = 0.433 mM, RSS 0.0371 (1.9 times larger), AIC -27.3 (ΔAIC = 7.8). Mixed: the ES-binding constant K_i' ran to its upper bound (500 mM), K_i = 0.107 mM, AIC -33.1, i.e. the mixed model reduces to the competitive one. The comparison favours the competitive mechanism, but with twelve single measurements and no inhibited plateau it cannot exclude a modest uncompetitive component, and the equal V_max of the competitive and mixed fits is a property of those models, not an independent observation.

Dataset	Tube	[ONPG]0 (mM)	A415 2 min	Converted at 2 min (%)	A415 10 min	Converted at 10 min (%)
Reference Part III (+IPTG)	A	2.2	0.198	2.6	0.966	12.6
Reference Part III (+IPTG)	B	1.1	0.065	1.7	0.294	7.7
Reference Part III (+IPTG)	C	0.55	0.045	2.4	0.187	9.8
Reference Part III (+IPTG)	D	0.275	0.029	3	0.099	10.3
Reference Part III (+IPTG)	E	0.138	0.02	4.2	0.069	14.4
Reference Part III (+IPTG)	F	0.0687	0.011	4.6	0.036	15
Reference Part III (-IPTG)	A	2.2	0.282	3.7	1.33	17.3
Reference Part III (-IPTG)	B	1.1	0.202	5.3	0.723	18.9

Dataset	Tube	[ONPG]0 (mM)	A415 2 min	Converted at 2 min (%)	A415 10 min	Converted at 10 min (%)
Reference Part III (-IPTG)	C	0.55	0.156	8.1	0.468	24.4
Reference Part III (-IPTG)	D	0.275	0.11	11.5	0.301	31.4
Reference Part III (-IPTG)	E	0.138	0.116	24.2	0.18	37.6
Reference Part III (-IPTG)	F	0.0687	0.047	19.6	0.09	37.6
Experimental Part III (+IPTG)	A	2.2	0.163	2.1	0.78	10.2
Experimental Part III (+IPTG)	B	1.1	0.269	7	1.11	28.9
Experimental Part III (+IPTG)	C	0.55	0.08	4.2	0.362	18.9
Experimental Part III (+IPTG)	D	0.275	0.041	4.3	0.188	19.6
Experimental Part III (-IPTG)	A	2.2	0.259	3.4	1.06	13.9
Experimental Part III (-IPTG)	B	1.1	0.224	5.8	1.06	27.8
Experimental Part III (-IPTG)	C	0.55	0.2	10.4	0.88	46
Experimental Part III (-IPTG)	D	0.275	0.202	21.1	0.764	79.8
Experimental Part III (-IPTG)	E	0.138	0.111	23.2	0.346	72.3

Table A14. Substrate converted by 2 and 10 min, from $A_{415}/(\varepsilon \times 2/3)$ relative to nominal [ONPG]. Reference uninhibited tubes D to F reached only 31 to 38% conversion by 10 min yet their rates fell to a few percent of the initial rate. Candidate explanations and the test each implies: substrate depletion (predicts a plateau at 100% conversion; observed about 38% in E and F); enzyme inactivation (excluded by tube A, same enzyme and session, linear to 10 min); late stopping of the first aliquot (lowers V_0 but cannot create a plateau); a lower-than-nominal effective ONPG concentration in the low tubes (would place the plateau at the true concentration; testable by reading the ONPG stock's own absorbance or by complete hydrolysis of an aliquot); a background absorbance unrelated to product (would appear as a non-zero intercept; the linear curves show intercepts of 0.00 to 0.06 A, too small to explain plateaus of 0.09 to 0.30 A). The experimental uninhibited D and E tubes, by contrast, reached 80% and 72% conversion and curved for the expected reason.

5. Secondary analysis: the experimental dataset

+IPTG tube B	Method	Km (mM)	Km,app (mM)	Vmax (umol/min)	Vmax,app (umol/min)	Vmax,app/Vmax	Ki (mM)	R2 (-I)	R2 (+I)	Tubes (-I / +I)
excluded	Lineweaver-Burk	0.156	3.087	0.421	0.792	1.88	0.024	0.95	0.969	ABCDE / ACD
excluded	Nonlinear MM	0.137	1.206	0.409	0.414	1.01	0.059	0.942	0.984	ABCDE / ACD
included	Lineweaver-Burk	0.156	38.16	0.421	9.042	21.5	0.002	0.95	0.945	ABCDE / ABCD
included	Nonlinear MM	0.137	0.783	0.409	0.458	1.12	0.097	0.942	0.541	ABCDE / ABCD

Table A15. Kinetics from our Part III printouts with the same rule and units (uninhibited E is a Tier-2 secant). With +IPTG tube B excluded, the nonlinear fits give K_m = 0.137 mM, V_max = 0.409 µmol min^{-1}, K_m,app = 1.206 mM, V_max,app = 0.414 (ratio 1.01), K_i = 0.059 mM: the textbook competitive pattern, with K_m near the literature 0.12 mM. The Lineweaver–Burk analysis of the same points is distorted by the three-point inhibited series (K_m,app 3.09 mM, V_max,app 1.88 × V_max) and, if tube B is included, becomes meaningless (K_m,app = 38 mM).

Condition	Tube	V0 reference (dA415/min)	V0 experimental (dA415/min)	Experimental / reference	Estimator, reference	Estimator, experimental	A415 at 10 min, reference	A415 at 10 min, experimental
Enzyme dilution (Part II)	A	0.136	0.1428	1.05	regression 2-6 min	regression 2-8 min	1.407	1.328
Enzyme dilution (Part II)	B	0.1041	0.085	0.82	regression 2-10 min	regression 2-10 min	1.084	0.86
Enzyme dilution (Part II)	C	0.082	0.0609	0.74	regression 2-6 min	regression 2-8 min	0.736	0.583
Enzyme dilution (Part II)	D	0.0243	0.0275	1.13	regression 2-6 min	regression 2-10 min	0.262	0.283
Enzyme dilution (Part II)	E	0.0229	0.0197	0.86	regression 2-10 min	regression 2-10 min	0.248	0.205
ONPG series, no inhibitor	A	0.132	0.1126	0.85	regression 2-10 min	regression 2-8 min	1.328	1.061
ONPG series, no inhibitor	B	0.101	0.1055	1.04	0-2 min secant (no initial linear portion)	regression 2-10 min	0.723	1.065
ONPG series, no inhibitor	C	0.078	0.0894	1.15	0-2 min secant (no initial linear portion)	regression 2-8 min	0.468	0.88
ONPG series, no inhibitor	D	0.055	0.0872	1.59	0-2 min secant (no initial linear portion)	regression 2-6 min	0.301	0.764
ONPG series, no inhibitor	E	0.058	0.0555	0.96	0-2 min secant (no initial linear portion)	0-2 min secant (no initial linear portion)	0.18	0.346
ONPG series, no inhibitor	F	0.0235	NA	NA	0-2 min secant (no initial linear portion)	NA	0.09	NA
ONPG series + IPTG	A	0.099	0.0771	0.78	0-2 min secant (no initial linear portion)	regression 2-10 min	0.966	0.78
ONPG series + IPTG	B	0.0328	0.121	3.69	regression 2-6 min	regression 2-8 min	0.294	1.106
ONPG series + IPTG	C	0.0208	0.041	1.98	regression 2-6 min	regression 2-6 min	0.187	0.362
ONPG series + IPTG	D	0.0087	0.0181	2.09	regression 2-10 min	regression 2-10 min	0.099	0.188

Condition	Tube	V0 reference (dA415/min)	V0 experimental (dA415/min)	Experimental / reference	Estimator, reference	Estimator, experimental	A415 at 10 min, reference	A415 at 10 min, experimental
ONPG series + IPTG	E	0.0062	NA	NA	regression 2-10 min	NA	0.069	NA
ONPG series + IPTG	F	0.0031	NA	NA	regression 2-10 min	NA	0.036	NA

Table A16. Tube-by-tube comparison. The standard tube agreed within 5% (enzyme dilution, 1.05) and 15% (substrate series, 0.85). Our uninhibited low-substrate tubes ran faster than the reference tubes (D: 1.59) because they stayed linear longer, and our inhibited tubes C and D ran 2.0- and 2.1-fold faster; +IPTG tube B ran 3.7-fold faster than the reference tube B and 1.57-fold faster than our own tube A.

+IPTG tube B. The observation is that velocity increased when substrate halved at fixed enzyme and inhibitor, which no kinetic mechanism allows; the tube's nominal composition is therefore unverifiable. Its first four points are as linear as any in the dataset (R^2 0.9999), so instrument error is unlikely. Hypotheses, in order of simplicity: IPTG omitted (the curve matches an uninhibited tube at high substrate), or a larger volume of ONPG or enzyme than intended. The 10-min value (1.106) is 46% of the increment the first four points predict and is consistent with the stop reagent added early or the aliquot mixed late; it is retained in the raw data and excluded from that tube's regression by the ordinary rule. The instrument clock stamps show the black set was read as a batch (04:20 to 04:33, order D, B, A, C), so read order carries no information about reaction order.

+IPTG tube E. Discarded before measurement; no value was estimated. The inhibited experimental series therefore rests on tubes A, C and D.

6. Sensitivity analysis

Estimator	Method	Km (mM)	Km,app (mM)	Vmax (umol/min)	Vmax,app (umol/min)	Km,app/Km	Vmax,app/Vmax	Ki (mM)	R2 (-I)	R2 (+I)	Physical (-I / +I)	Primary
A: all five points for every curve	Lineweaver-Burk	3.405	1.317	0.849	0.214	0.39	0.25	-0.747	0.965	0.988	TRUE / TRUE	
A: all five points for every curve	Nonlinear MM	20.9	NA	4.744	NA	NA	NA	NA	0.995	0.951	TRUE / FALSE	
O: two-tier rule, secant corrected for the t = 0 offset	Lineweaver-Burk	0.555	1.725	0.579	0.277	3.11	0.48	0.217	0.916	0.988	TRUE / TRUE	
O: two-tier rule, secant corrected for the t = 0 offset	Nonlinear MM	0.526	NA	0.528	NA	NA	NA	NA	0.924	0.971	TRUE / FALSE	
P: two-tier rule (primary)	Lineweaver-Burk	0.314	1.757	0.477	0.282	5.6	0.59	0.1	0.928	0.988	TRUE / TRUE	yes
P: two-tier rule (primary)	Nonlinear MM	0.363	NA	0.494	NA	NA	NA	NA	0.919	0.962	TRUE / FALSE	
R: initial-window regression only	Lineweaver-Burk	-3.48	1.722	-0.912	0.276	-0.49	-0.3	-0.307	0.99	0.989	FALSE / TRUE	
R: initial-window regression only	Nonlinear MM	3.379	NA	1.147	NA	NA	NA	NA	0.998	0.972	TRUE / FALSE	
S: 0-2 min secant for every curve	Lineweaver-Burk	0.322	0.62	0.487	0.188	1.93	0.39	0.495	0.928	0.984	TRUE / TRUE	
S: 0-2 min secant for every curve	Nonlinear MM	0.436	NA	0.537	NA	NA	NA	NA	0.908	0.954	TRUE / FALSE	
P: two-tier rule (primary)	Global nonlinear competitive	0.435	2.297	0.529	0.529	5.28	1	0.107	0.919	NA	TRUE / NA	

Table A17. Kinetic parameters from the reference dataset under five V_0 estimators. Regression on every curve (R) gives a negative uninhibited Lineweaver–Burk intercept, so K_m and V_max are undefined; all five points (A) give K_m = 3.4 mM; the 0 to 2 min secant for every curve (S) gives K_m = 0.32 mM but collapses K_m,app to 0.62 mM and raises K_i to 0.50 mM, because the secant inflates the low-signal inhibited tubes (positive intercepts of 0.006 to 0.014 A make the 2-min reading a poor origin-anchored estimate); the offset-corrected rule (O) raises K_m to 0.55 mM (K_m,app 1.73 mM, ratio 3.1) and K_i to 0.22 mM. The two-tier rule (P) is the estimator that follows the manual's instruction (initial linear portion where one exists, earliest interval where none exists) and gives physically admissible parameters for both series; it was not chosen for the values it produces. Conclusions that survive every estimator: velocity saturates with substrate, and IPTG slows the enzyme progressively more as substrate falls. Conclusions that depend on the estimator: the numerical K_m (0.31 to 0.55 mM among the rule-based estimators and the global fit) and, through it, K_i (0.10 to 0.22 mM among those estimators; 0.50 mM under the uniform 0 to 2 min secant, which is biased upward for the inhibited tubes).

7. Limitations

Single measurements per condition (no replicate variance; quoted uncertainties are regression standard errors); manual 2-min sampling that cannot resolve the initial rate when it falls within the first interval, which affects five of six uninhibited reference tubes; a substrate range whose highest concentration (2.20 mM) is near the uninhibited K_m estimate and below the apparent K_m with IPTG, so the inhibited plateau was never reached and V_max,app is extrapolated; the origin assumption of the secant estimates, bounded but not removed by the offset test; read-out resolution of 0.001 A relative to increments of 0.005 A in the most dilute inhibited tubes; nominal ONPG concentrations that the plateaus of the low-substrate reference tubes suggest may have been lower than stated; room temperature uncontrolled; the experimental inhibited series reduced to three tubes by the loss of E and the anomaly in B; literature values measured for the purified enzyme in TES buffer with 145 mM NaCl at pH 7.0 and 25 °C by a continuous assay at 420 nm, so exact agreement is not expected.

8. Reproducibility

```
Rscript R/run_all.R
```

regenerates `data/processed/`, `tables/`, `figures/` and the three report sources from the raw files (R 4.3.3; tidyverse, broom, patchwork, scales, here; ragg optional; XeLaTeX for the PDF versions). Analysis settings live in `R/00_setup.R` (tolerances of the V_0 rule, the excluded experimental tube, the K_m bound). Session information of the last run is in `report/sessionInfo.txt`. Code and data: public repository https://github.com/almokhtar8-stack/exp2. The course's original files and the instructor's e-mail are kept locally for the cross-check in `01_import_clean.R` but are not redistributed (permission not obtained); the scripts run without them.

9. References

See the reference list in `experiment_2_submission.md`. Literature comparison values: Juers et al. (2009) Protein Sci. 18, 1281-1292 (Table II, native enzyme: k_cat 620 s^{-1}, K_m 0.12 mM for oNPG; Table IV: K_i 0.08 mM for IPTG at 0.01 mM Mg^{2+} and ~150 mM NaCl; TES buffer, pH 7.0, 25 °C; ε(oNP) 2.65 mM^{-1} cm^{-1} at 420 nm, pH 7.0). The classical alkaline-stop assay and its extinction coefficient: Miller (1972) Experiments in Molecular Genetics, pp 352-355. Instrument specification: Thermo Fisher Scientific, GENESYS 10S Series UV-Visible Spectrophotometers brochure (photometric linear range up to 3.5 A at 260 nm; accuracy ± 0.005 A at 1.0 A).